

## **Self-Assessment**

Before starting the Computer Science program at SNHU I had very limited knowledge of coding and general CS concepts. Upon completion of the CS program, however, I built skills in areas such as cyber and network security and learned about modern trends such as AI advancement. I had engaged with concepts that I had never even known about beforehand, including running JUnit tests, and performed tasks I didn't think were accessible to me, such as building small functional mobile apps.

My work in SNHU has required me to become familiar with meeting the needs of end users and interpreting feedback from stake holders. Many assignments required me to not only read and act on mock emails written from fake managers, but also to use my own critical thinking skills to determine the kind of features end users may require to successfully use my applications. Now when I build projects, I have no trouble considering how it may be intentionally broken by hackers, common uses and mistakes of end users, and the desired output of stakeholders and product managers.

My skills are strongest in data structures and algorithms, most likely due to a natural curiosity and enjoyment for this section of CS. Even though the quality and validation of datasets and the visualization and interpretation requirements come naturally to me, I don't have trouble finding the solutions to many programming and engineering questions and continue to practice a methodical approach to guarantee I build strong and thorough projects.

My projects fit together on my portfolio because they show how far I've come as a programmer and illustrate the different ways I can answer one question. The quality jump from Paint1 to Paint2 helps to demonstrate my growth as a developer. The way I answered common

questions between Rapify and Wrapify shows my diverse skillset and the ways in which I can achieve similar results using different methods. Wrapify especially shows how I consider end user's needs and create something that benefits many types of users.

## **Paint2**

Paint2 is an enhanced version of one of my very first CS projects, Paint1. At the time of creating Paint1 back in 2022 I was brand new to coding as a concept and was using Java for the first time. The original file technically does not work in its untouched state due to the calling of some incorrect variables, but upon correction, the program will calculate the gallons of paint needed to cover one wall.

Paint2 expands on this functionality, allowing users to input any number of walls with varying widths. I went even further to include the ceiling as a paintable object and allowed users to input door frames and windows to remove from the final calculation.

I included this artifact because it is a great example of my improvement throughout the years that I have studied CS, showing that I went from making small projects that hardly worked to much more complicated ones that run smoothly and efficiently. It may not be the most impressive project, but it shows my commitment to learning and improving existing skills.

As I was creating and improving upon Paint2, I learned about the various methods of error handling. I have been incredibly used to using if-else statements to ensure that my program only runs when certain criteria is met, but I hit a wall when string input was inserted. My desired outcome was for the program to tell the user the correct input, and prompt them again, but instead my program would just stop running. That's when I did some research on how other developers handled errors in their work and came across exceptions. After reading an article

about how to write exceptions, I was able to pair this process with my pre-existing if-else statements, and suddenly my program was running exactly as I'd hoped.

With my enhancements I feel I met a few course outcomes, namely: *“Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.”* Due to my more advanced artifact that now meets the needs of multiple kinds of users. I also met: *“Develop a security mindset that anticipates adversarial exploits in software architecture and designs to expose potential vulnerabilities, mitigate design flaws, and ensure privacy and enhanced security of data and resources.”* Since I included more robust error handling and my program does a good job of mitigating incorrect input.

## **Rapify**

Rapify was a project I had been considering creating for a few months, and I'd even gone as far as cleaning as much as I could from the dataset that I had acquired. Rapify is an R project that takes user data from Spotify to output the most popular songs and artists for that user.

I believe this artifact shows my diverse skillset and would look especially appealing to recruiters that are looking for a more data-driven project. I demonstrated my ability to clean, query, and visualize a dataset and interpret results for users.

When enhancing this project, I went into it with a different mindset, thinking I'd be able to use machine learning to get all the information I wanted. However, since this isn't numerical data, machine learning wound up being a very difficult choice to implement. Instead, I took this as my opportunity to try out many different R libraries to see what kind of features I could use to build a project similar to what I had wanted to make originally. After playing with the data, I

realized that the “skipped” column might be pretty easily turned into numeric data, and built a model that somewhat accurately predicts the likelihood of a song being skipped.

The course outcome I met with this project includes: “*Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices.*” In which I had to choose implementing my original functionality over whatever results I would’ve pulled from machine learning. I also met: “*Design, develop, and deliver professional-quality oral, written, and visual communications that are coherent, technically sound, and appropriately adapted to specific audiences and contexts.*” Since I had to design the output to be easily legible to all types of users.

## **Wrapify**

Wrapify is the project I am the proudest of. Wrapify was made prior to Rapify, but does the same thing using python instead of R. The difference between Wrapify and Rapify, however, is that Wrapify will eventually be open to use by the public, in which anybody will be able to input their streaming history and get the results back that tell them all about their personal tastes. For the sake of this project my only goal was to complete a lot of base functionality, but I am hoping to continue working on this project well into the future and make it accessible even to other listening apps like Apple Music and Band Camp.

This artifact is meant to show what I am passionate about, and the various ways in which I can build, clean, query, and visualize data from a large dataset. This project shows my ability to break a dataset up into readable chunks, and determine what information is useful and how it can be used to find answers.

When enhancing this artifact, one of the main concerns I had was making sure the final output could be read by anyone. I wanted to ensure that everything had two ways it could be read, as a table and as a chart. The table was easy enough to output, but it took me quite a few tries to figure out how to graph the number of times string input can be found in the dataset. The Seaborn library really helped me with this, and I wound up relying heavily on their pre-built bar chart.

The course outcomes I feel I met include: “*Employ strategies for building collaborative environments that enable diverse audiences to support organizational decision making in the field of computer science*” Since this project is meant to be shared with many people, and I want their feedback it’s a great opportunity for a diverse audience to change the way their data can be used. The other outcome I feel I met includes: “*Demonstrate an ability to use well-founded and innovative techniques, skills, and tools in computing practices for the purpose of implementing computer solutions that deliver value and accomplish industry-specific goals.*”